THE VIRTUAL MEET EXPERIENCE

2018 - 2019

HS VIRTUAL CHALLENGE

STATE QUALIFIERS' MEET

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2018-2019 Virtual Challenge State Qualifiers' Meet – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.x**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is 78_9 plus 45_{10} ?

A. 11022_3 B. 421_5 C. 311_6 D. 1010101_2 E. All are equivalent.

QUESTION 2

What is output by the code to the right?

A. 9 B. 4 C. 7 D. 6

E. There is no output due to a syntax error.

```
out.println(5+4/2);
```

QUESTION 3

What is output by the code to the right?

A. .2f B. 5.67 C. 5.666 D. 5.7

E. There is no output due to a syntax error.

```
out.printf(".2f",5.666);
```

QUESTION 4

What is output by the code to the right?

A. somebodyoncetoldme

B. somebody

C. Somebody

D. Somebodyoncetoldme

E. There is no output due to a syntax error.

```
String s="Somebody";
s=s.concat("oncetoldme");
out.println(s);
```

QUESTION 5

What is output by the code to the right?

A. false B. true C. 0 D. 1

E. There is no output due to a syntax error.

```
out.println(true && true ^ true);
```

QUESTION 6

What is output by the code to the right?

A. 6 B. 5.0 C. 6.0 D. 5

E. There is no output due to an error.

```
double j=5.6;
int h=Math.round(j);
out.println(h);
```

QUESTION 7

What is output by the code to the right?

A. 7 B. 14 C. 28 D. 0

E. There is no output due to an error.

```
int h=7;
h+=h;
out.println(h);
```

QUESTION 8

What is output by the code to the right?

A. 2 B. 1

C. 12 D. 22

E. 212

```
int x=7;
if(x<=9 ||(++x)>9)
  System.out.print(2);
if(x==8)
  System.out.print(1);
System.out.println(2);
```

QUESTION 9

What is output by the code to the right?

A. 123456789
B. 2345678910
C. 23456789
D. 12345678910
E. 0123456789

```
for(int y=1;y<10;++y)
  System.out.print(y);
```

QUESTION 10

What is output by the code to the right?

A. 55
B. 45
C. 100
D. 50
E. 0

```
int s=0;
for(int y=0;y<10;y++)
  for(int x=0;x<y;x++)
     s++;
out.println(s);
```

QUESTION 11

Which of the following lines could read in the following dataset without error?

```
Hello There. "General Kenobi"
```

A. file.next()
B. file.nextInt()
C. file.nextLine()
D. file.nextDouble()
E. All of the above lines can read the data.

QUESTION 12

What is output by the code to the right?

A. 5
B. 4
C. 0
D. 2
E. 3

```
int[] y={1,2,3,4,5,0};
int k=y[0];
for(int x=0;x<22;x++)
  k=y[k];
System.out.println(k);
```

QUESTION 13

What is output by the code to the right?

A. yes B. true C. false D. no E. 5

```
out.println(100%3==0?false:true);
```

QUESTION 14

What is output by the code to the right?

A. 2
B. 16
C. 64
D. 32
E. There is no output due to a syntax error.

```
out.println(Float.SIZE);
```

QUESTION 15

What is output by the code to the right?

A. 0
B. 1
C. 2
D. 3
E. There is no output due to a runtime error.

```
out.println(14 % 4 % 3 % 8);
```

QUESTION 16

What could fill **<1*>** so that Test works as intended??

A. `throws`

B. `extends`

C. `implements`

D. `is`

E. B or C

QUESTION 17

What is output by the line marked `//1` in the code to the right?

A. `64`

B. `2`

C. `3`

D. `1`

E. There is no output due to an error.

QUESTION 18

What is output by the line marked `//2` in the code to the right?

A. `nameless`

B. `cnameless`

C. `chnameless`

D. `chinameless`

E. There is no output due to a syntax error.

QUESTION 19

Method `sqr` is an example of what?

A. Inheritance

B. Overloading

C. Overriding

D. All of the above.

E. None of the above.

```
interface face {
  public String ss(String s, int y);
  public int sqr(int j);
}

public class Test <1*> face{

  public String n;

  public Test() {
     n = "nameless";
  }

  public String ss(String s, int y) {
    return
      s.substring(0, sqr(y - 1)) + n;
  }

  public int sqr(int j) {
    return (int)(Math.pow(j, 2));
  }

  public int sqr(int j , String s) {
    return (int)
     (Math.pow(j , s.length() - 1));
  }
}


//////////////client code///////////
Test c = new Test();
out.println(c.sqr(c.n.length())); //1
out.println(
     c.ss("chicken fingers", 2)); //2
```

QUESTION 20

What is output by the code to the right?

A. `oe_1177n3-S221G60a557564CigeHol-13992nn62`

B. `Col25n33o-iH191n--n3722ne1253_CnSGaC1ngeH31`

C. `_-SianHol-onCGnege`

D. `12l-225oa3g-i3195-one_CSG513nne1122-3eH337310`

E. There is no output

```
String h="Hole-in-one_CSGang";
for(int y=0;y<10;y++) {
int k=((((y+2)*3)-1)*4+1)%h.length();
h=h.substring(k+1)+
        h.charAt(k)+h.substring(0,k);
h=h.substring(h.length()/2)+
         h.substring(0,h.length()/2);
h+=h.charAt(k)+k;
}
out.println(h);
```

QUESTION 21

What is output by the code to the right?

A. `1998` B. `2000` C. `1999` D. `2002` E. `2001`

```
out.println(60<<5|45&234>>3|78);
```

QUESTION 22

What output by the code to the right ?

A. 673 B. 674 C. 675 D. 676 E. 677

```
int h=99;
String s=Integer.toString(h,4);
h=Integer.parseInt(s,7);
s=Integer.toString(h,8);
h=Integer.decode(s);
out.println(h);
```

QUESTION 23

What is output by the code to the right?

A. 12.9 B. 12.0 C. 12 D. 13.0

E. There is no output due to a compile error.

```
Double d=3;
d*=4.3;
System.out.println(d);
```

QUESTION 24

What is output by the line marked //1 in the client code to the right?

A. 46

B. 47

C. 48

D. 49

E. 50

QUESTION 25

How many calls are made by the line marked //2 in the client code to the right?

A. 109

B. 108

C. 107

D. 106

E. 110

QUESTION 26

What is output by the line marked //2 in the client code to the right?

A. 134

B. 135

C. 136

D. 137

E. 138

```
public static int recur(int y){
  if(y<0)
    return 1;
  return recur(y-1)+recur(y-2)+y;
}

//////////////////////////////////////
System.out.println(recur(5)); //1
System.out.println(recur(7)); //2
```

QUESTION 27

What is the Big O value for a O(n) algorithm for a list of length 8,000 items, if the algorithm takes 1 second to process a list with a length of 2,000?

A. 2 seconds

B. 4 seconds

C. 8 seconds

D. 16 seconds

E. None of the above.

QUESTION 28

What is output by the code to the right?

A. 5

B. 7

C. 8

D. 9

E. There is no output due to a runtime error.

```
ArrayList<Integer> a;
a=new ArrayList<Integer>();
for(int y=1;y<10;y++){
  a.add(y);
  if(y%2==0)
     a.remove(0);
}
System.out.println(a.get(4));
```

QUESTION 29

What is output by the code to the right?

A. [0] B. [, , , 0]

C. [, 0] D. []

E. There is no output due to an error.

```
String[]s="abc0".split("[a-z]");
out.println(Arrays.toString(s));
```

QUESTION 30

Which of the following Java statements will compile and is equivalent to the mathematical formula to the right?

A. long E=m*c*c;

B. long E=m*Math.pow(c);

C. long E=m*(c^2);

D. All of the above.

E. None of the above.

$E=mc^2$

QUESTION 31

What is the sum of the ASCII values of all the characters contained in String s?

A. 897 B. 898 C. 899 D. 900 E. 901

```
String a="IRunTheNumbers";
String s="LikeKevin";
```

QUESTION 32

Which of the following is equivalent to the binary value 1101001?

A. 253_6

B. 1221_4

C. 10220_3

D. 210_7

E. All are equivalent.

QUESTION 33

Which of the following is not a valid Java identifier?

A. SomebodytoLove!

B. TheVolume211

C. Lucky#9

D. $$$$

E. More than one of these.

QUESTION 34

Which of the following import statements will allow code to implement the `Rectangle` class?

A. `import java.math.*;`

B. `import java.io.*;`

C. `import java.util.*;`

D. `import java.awt.*;`

E. `import java.lang.*;`

QUESTION 35

What is last value dequeued in the queue psuedocode to the right?

A. `212`

B. `17`

C. `9`

D. `8`

E. `5`

```
Push 5
Pop X
Push 5
Push 7
Push 212
Push 17
Push 8
Push 9
Pop X
Pop X
Push 9
Pop X
```

QUESTION 36

What is the decimal equivalent of the two's complement value, 11101010_2?

A. -22

B. -23

C. -21

D. -20

E. None of the above.

QUESTION 37

Which of the following is the cheapest path through the graph to the right?

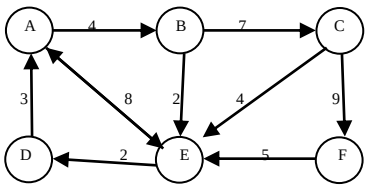
A. ABCFEDA

B. ABEDA

C. AEDA

D. ABEA

E. ABCEDA



QUESTION 38

If the following items were added to a binary search tree in the order show below, what would a post order traversal display?

17 212 -3 89 2 12 3 27 4 0 90 2 4 7

QUESTION 39

Using the tree from the above question, how many leaves exist?

QUESTION 40

What is the height of the tree created in question 38?

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2018-2019
Virtual Challenge State Qualifiers' Meet
Computer Science - KEY

1) A
2) C
3) A
4) D
5) A
6) E
7) B
8) D
9) A
10) B

11) C
12) A
13) B
14) D
15) C
16) C
17) A
18) B
19) D
20) A

21) C
22) B
23) E
24) B
25) A
26) A
27) B
28) D
29) B
30) A

31) B
32) E
33) E
34) D
35) A
36) A
37) B
38) [2,0,4,7,4,3,12,2,-3,27,90,89,212,17]
39) 5
40) 6

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors.**
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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